Are NBFNet's Path Explanations Actually Faithful? A Deletion, Edit and Neuromodulation Benchmark on a Frozen Knowledge Graph Reasoner

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Abstract

NBFNet explains a link prediction by reading paths out of the gradient of its own score with respect to each edge, and presents those paths as the provenance of the answer. I test that claim on a frozen public checkpoint by intervening on the graph instead of inspecting it: four graphs, two of them held out, 800 correctly answered queries in a fixed random order, 15,447 single-path deletions, joint deletions, random controls, edits that reroute a final hop, and a dose-response test that turns whole relation types down like a neuromodulator. The audit found a bug. The released explanation code differentiates a version of the model whose messages run backwards, so its gradient is off by as much as the effect it predicts (a median error of 100%) and has the wrong sign on 21 of 50 random directions; a small fix brings the error to 0.7%. With or without the fix, the selected edges are load-bearing: deleting the shortlist removes between 0.259 and 0.858 of the gold tail's probability, against at most 0.007 for as many random edges. The order inside the shortlist carries little: Kendall's τ against measured effect is between -0.009 and 0.076 as released. The fix adds about 0.1, and integrated gradients over the same candidates reach 0.489 on CoDEx-S and 0.445 on NELL23k, still far from 1. Turning a relation down shows why a slope misleads: the response is often a switch, with between 0.030 and 0.290 of a channel's effect present at half gain, where a straight line gives 0.5. The ULTRA paper's own checkpoint reproduces its reported accuracy and gives the same picture. Whether the shortlist is many explanations or one edge written out many times depends on the graph, and the output never says which.

1. Introduction

NBFNet [14] represents a pair of entities by the paths between them, and it explains its own predictions from the same computation. Each edge is given a weight, the model is run, and the gradient of the score with respect to those weights is read off. A beam search over those gradients returns the top paths, and the paper presents them as the paths behind the answer. That the explanation is part of the computation, rather than fitted on top of it, is what makes it attractive and what makes it worth checking.

A gradient tells you what happens if an edge changes a little. Whether the model relied on the edge is a question about taking it away, and the two can disagree. In earlier work on siRNA efficacy models I found saliency maps pointing at sequence positions that mattered far less than the attribution implied, and only perturbing the input showed it [7]. The case for paths over embeddings is provenance: a nearest neighbour in vector space cannot tell you why, and a path can. That only works if the path shown is the one doing the work. A wrong explanation is worse than none, because a reader who checks the path and finds it sound will conclude the prediction is sound.

Where this sits in the course. Topic 8 asked the same question in a different place. Hase and colleagues found that where causal tracing says a fact lives barely predicts where an edit succeeds, with a correlation near -0.13 [6]. MQuAKE found an editor recalling 96.2% of the facts it wrote while multi-hop accuracy on questions that depend on them fell from 40.5% to 7.0% [13]. The Mirage paper found that once a model has to generate freely, mean reliability over six editors falls from 96.8% to 38.5% [11]. In each case a method passed an inspection test and failed an intervention test. This project runs that argument on a graph reasoner, where the intervention is literally a knowledge edit and, unlike weight editing, there is no question about whether it took. The model under test is a Topic 2 method, NBFNet inside the ULTRA foundation model, and the research track asks for a course method to be improved and compared one variable at a time. The variables here are the direction of message passing inside the explanation, the attribution rule (gradient or integrated gradients), the checkpoint, and whether the graph was seen in training. Random deletion is the ablation,

and the reverse-fact case in Section 7.7 is the error analysis, and nDCG, the retrieval measure from the course, scores the rankings.

Contributions. (1) A deletion and edit audit of NBFNet's released explanation code, against a frozen public checkpoint, with no training, on a benchmark of four graphs sized by a reliability analysis rather than by budget. (2) A bug in that code, shared with NBFNet's own PyTorch Geometric release: the explanation differentiates the model with every message reversed. I verify it three ways and fix it. (3) The finding that the selected edges are load-bearing on every graph while the order inside the shortlist carries little, before and after the fix, and on the ULTRA paper's own checkpoint. (4) Two measured improvements to that order, the direction fix and integrated gradients, neither of which makes it faithful. (5) A relation-level dose response, after Bull *et al.*'s account of neuromodulation [1], that shows the model's channels behave as switches a local slope cannot see. (6) The finding that whether the shortlist is many explanations or one edge depends on the graph, and that the output does not reveal which.

2. Background

NBFNet generalises Bellman-Ford into a neural network. To score $(h, r, ?)$ it initialises the head with a representation of the relation, then runs L rounds of relational message passing, so that a node's state summarises the paths of length at most L that reach it from h. I run ULTRA [5], whose entity-level reasoner is NBFNet unchanged, in a class the repository calls `EntityNBFNet`; ULTRA adds a second NBFNet over a graph of relations that conditions the first. That relation-level pass turns the query relation into one vector per relation type, and in the entity graph every message along an edge of type c is multiplied by the vector for c. ULTRA ships pre-trained checkpoints in its repository, so the model being explained is a published artefact.

The explanation is not my reimplementation. `BaseNBFNet.visualize` runs the forward pass with a per-edge weight vector marked for automatic differentiation, takes $\partial s / \partial w_e$ for every edge at every layer, beam-searches the highest-scoring paths under those gradients, and returns the top k. I call that function and audit what it returns; Section 5 shows that it differentiates a different model from the one it explains.

3. Related work

Faithfulness. ERASER [4] made the two standard measures concrete: comprehensiveness, the drop when the highlighted part is removed, and sufficiency, what survives when only it is kept. I use both, on probabilities so that queries can be compared. **Graph explanations.** GNNExplainer [12] set the pattern of learning a mask that preserves the prediction; most work since produces explanations rather than testing them, usually against synthetic graphs with a planted motif, which cannot say whether a method matches what a model depends on in a real benchmark. A*Net [15] scales path-based reasoning further, which makes the question of what its paths are worth more pressing. **Instability.** Méroué and colleagues report that models with the same MRR return different top-K predictions across seeds [9], so an explanation from one run deserves error bars. **Neuromodulation.** In nervous systems a few modulators, broadcast by volume transmission rather than wired point to point, change what a fixed circuit does without changing its wiring [8]. Bull and colleagues [1] built this into a recurrent network in which a low-dimensional modulator state multiplies the effective connectivity, $W_{\text{eff}} = W_0 + \sum_k T_k n_k$, and showed that it separates slow context from fast content. The dose response of such systems need not be monotone: dopamine's effect on working memory follows an inverted U [2], so a small change in level says little about a large one. The same concern about gradients, that a flat or saturated local response can hide a large global effect, motivates integrated gradients [10].

4. Method

Write $s_G(h, r, t)$ for the model's score on graph G and $p_G(t \mid h, r)$ for the softmax of that score over all entities. Every comparison is on p, because the logit has no fixed scale across queries, and the gold tail's rank is reported beside it.

Conditions. *A, gradient*: call `visualize` and take its top paths in its order. *B, single-path deletion*: delete each of the top 20 paths from the graph one at a time and measure the drop, which gives the ordering the gradient should have produced. *C, set level*: delete the top 5 jointly (comprehensiveness), and separately keep only their edges (sufficiency). *D, rewire*: point the final hop of the top path, the edge arriving at the predicted tail, at the runner-up entity instead, moving its stored inverse with it, and ask how much the old answer loses and the new one gains.

E, neuromodulatory channels. ULTRA has the factorization of Bull *et al.* [1] built in. Its relation-level pass runs once per query and its output multiplies every message of a relation type, so each relation type is a channel that reaches all of its edges at once. I give each channel c a gain g_c that scales every message along relation c and its inverse, with $g = 1$ the unmodified model. The gradient $\partial p / \partial g_c$ at $g = 1$, one backward pass for all channels, is the local slope. It is the gradient of p with respect to one weight shared by all of a channel's edges in every layer, so it is the edge-level signal NBFNet's explanation reads, pooled over the channel, and it is taken through the corrected path of Section 5. The dose response is p as g_c is turned down to 0.75, 0.5, 0.25 and 0. It is measured for every channel the top 5 paths use, for the 3 other channels the gradient ranks highest, and, at

0.5 and 0, for 3 random channels as a control; then all explained channels are turned down together. A gain of zero on a set of edges is the same computation as deleting them, so each query first zeroes the gain on exactly the shortlist's edges and checks that this reproduces the deletion recorded in condition C; the largest gap over 550 queries is 0. Turning a channel down is a knowledge edit at the level of a relation rather than a fact.

Interventions are edits. A deletion removes edge columns from `edge_index`, so the model runs on a graph that genuinely lacks those facts. The relation graph that conditions ULTRA is held fixed, so that removing a handful of edges cannot also move the relation representations and mix a second-order change into the measured drop.

Controls. *Random edges*: delete as many uniformly drawn edges as the shortlist contains. *The bottom of the pool*: the method's own lowest-ranked candidates are a within-query null for its ranking. *Tail degree*: a tail with one edge would make any deletion look decisive, so degree is recorded and the central result repeated on tails of degree four or more. *Reverse facts*: some graphs keep symmetric relations, so a query's reverse can already sit in the training graph and make it a one-hop lookup. Each query records whether it does. I added this last control after finding the problem in a worked example rather than by design (Section 7.7).

5. The explanation differentiates a different model

ULTRA, like NBFNet's PyTorch Geometric release, runs its layers with a fused sparse kernel, `rspmm`, whenever it can. When the edge weights need a gradient, which is exactly when `visualize` runs, each layer falls back to PyTorch Geometric's generic message passing. The two disagree on direction. The fused kernel sends the message on column e from `edge_index[1,e]` to `edge_index[0,e]`, and the generic path sends it from `edge_index[0,e]` to `edge_index[1,e]`. Training and inference use the kernel, so that is the deployed model. The explanation differentiates the other one: the same weights with every message running backwards.

I found this because integrated gradients did not add up. Its attributions must sum to the change in p between the empty graph and the full one. With the released gradient they missed it by a median of 63% of that change on the first 30 CoDEx-S queries, against 15% with the corrected one. Three checks on 50 benchmark queries from CoDEx-S and NELL23k then settle it (Figure 1). First, the function the explanation differentiates scores entities up to 10.5 logits away from the deployed model; run on the graph with every edge reversed, the same code reproduces the deployed scores to within 6×10^{-6}. Second, along a random direction in edge-weight space, the corrected gradient predicts the deployed model's change in p to a median of 0.7%, and the released one misses by a median of 100% and has the wrong sign on 21 of the 50 directions. The corrected error is not zero because a finite difference in single precision is itself approximate: with a step of 10^{-3} instead of 10^{-4} it is 0.2%. Third, for each query I take the explained relation channel on which the two gradients disagree most and measure the slope of p against that channel's gain at 1. The released gradient has the wrong sign on 16 of these 50 channels and the corrected one on 3.

The fix is to run `visualize` on the reversed graph and map each hop back to its original edge column. I keep both versions. The released one is what anyone calling the function gets, so the audit below reports it as shipped, and the path benchmark is repeated with the corrected one (Figure 4). The channel gradients and integrated gradients use the corrected path only.

6. The benchmark

Model and graphs. ULTRA's `ultra_4g` checkpoint, 168,705 parameters, frozen, on two CPU cores. It was pretrained on FB15k-237, WN18RR, CoDEx-M and NELL-995. WN18RR and FB15k-237 [3] are therefore in its mixture; CoDEx-S and NELL23k are not, although each comes from the same source as a graph that was. Table 1 gives the graphs and the reproduction, which matters first: if the checkpoint did not reproduce, the thing being explained would not be the published model.

Queries. The test triples are not in the message-passing graph, so an explanation cannot point at the answer itself. They are taken in a fixed random order and kept when the model ranks the gold tail in its top ten, because asking what a prediction rests on only makes sense when the prediction is right. For each query the beam returns up to 20 candidate paths; all are deleted individually, and the top 5 form the shortlist.

How many queries is enough? The model is deterministic, so the only source of variation is which queries are drawn, and every interval below is a 95% bootstrap over queries. To size the sample I computed each headline estimate on the first n queries for growing n, which is a valid sample because the order was fixed in advance (Figure 2). The estimates settle and the bands narrow roughly as $1/\sqrt{n}$. The same analysis shows what a small study would have risked. On WN18RR the disjoint blocks of five queries give a "top path beats chance" score anywhere from -0.51 to 0.33, and the running estimate still had the opposite sign at 50 queries; at full size it is -0.028, pinned to within ± 0.052. Across random splits of the WN18RR queries into two halves, the two halves' estimates of the shortlist deletion effect differ by at most 0.086 in 95% of splits.

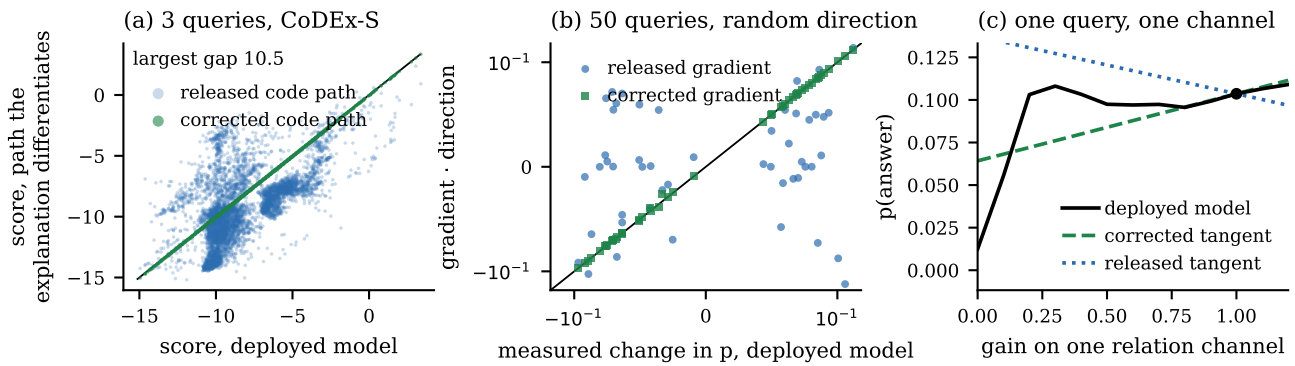


Figure 1. The released explanation code differentiates a model with its messages reversed. (a) Every entity's score from the code path the explanation differentiates, against the deployed model's score: as released (blue) and on the reversed graph (green). (b) Along one random direction in edge-weight space per query, the deployed model's measured change in p, by a central difference, against each gradient's prediction; the line is exact agreement. (c) One query and one relation channel: p as the channel's gain runs from 0 to 1.2, with each gradient's tangent at gain 1.

Table 1. The four graphs. MRR and Hits@k are filtered, both directions scored, with ULTRA's own evaluation code, on the number of test triples shown. "Paper" is the MRR Galkin et al. report for ULTRA pre-trained on three graphs [5]. The `ultra_4g` checkpoint used throughout adds NELL-995 to that mixture, so close agreement rather than identity is expected; the "3g" column is that paper's own checkpoint, run here on the full test set. Explained queries are the test triples the frozen model ranks in its top ten, taken in a fixed random order set before any measurement.

graph	entities	relations	edges	`ultra_4g` saw it	test scored	MRR	H@10	paper	3g	queries	scanned
WN18RR	40,943	11	173,670	yes	3134	0.484	0.617	0.480	n/a	200	288
FB15k-237	14,541	237	544,230	yes	2000	0.371	0.558	0.368	n/a	200	400
CoDEx-S	2,034	42	65,776	held out	1828	0.464	0.665	0.472	0.472	200	274
NELL23k	22,925	200	50,890	held out	4952	0.258	0.431	0.239	0.241	200	396

7. Results

7.1. The selected edges are load-bearing

The result that goes the method's way comes first, and it holds everywhere. Deleting the shortlist removes 0.858 of the gold tail's probability on WN18RR, 0.259 on FB15k-237, 0.427 on CoDEx-S and 0.368 on NELL23k (Table 2). Deleting the same number of random edges removes nothing measurable on any graph. The gradient is pointing at part of the graph the model depends on, and that is not a given.

Keeping only the shortlist does not give a clean answer in the other direction. With a handful of edges left, most rival entities are cut off too, so the gold tail can gain probability just because it has fewer competitors. It does on 8.0% of WN18RR queries, 36.5% of FB15k-237, 81.0% of CoDEx-S and 12.0% of NELL23k. I report the retained share for completeness and draw no conclusion about sufficiency from it. For a model that takes a softmax over every entity, a deletion test of necessity is well posed and a keep-only test is not.

7.2. As released, the order is close to meaningless

A faithful ranking would make Kendall's τ between the method's order and the measured drop close to 1. It is -0.009 on WN18RR, 0.048 on FB15k-237, 0.076 on CoDEx-S and 0.041 on NELL23k. Figure 3 shows it directly: walking down the method's own list, the effect of deleting each path stays flat. The plainest version of the question is how often the path ranked first is actually the most damaging one to delete. On WN18RR it is 0.253 against a chance rate of 0.280, no better than a random pick; on the other graphs it is at or modestly above chance. This is the shape of Hase et al.'s result one domain over [6]: a quantity read from the model's internals that correlates with what the model does is not thereby a measurement of what it depends on.

7.3. Fixing the direction helps the order, a little

Repeating the benchmark with the corrected explanation leaves the set-level result where it was (Figure 4): its shortlist removes 0.461 of the gold tail's probability on CoDEx-S and 0.394 on NELL23k, against 0.427 and 0.368 as released. The order improves. On the same queries Kendall's τ rises from 0.076 to 0.185 on CoDEx-S, a gain of 0.109 [0.064, 0.155],

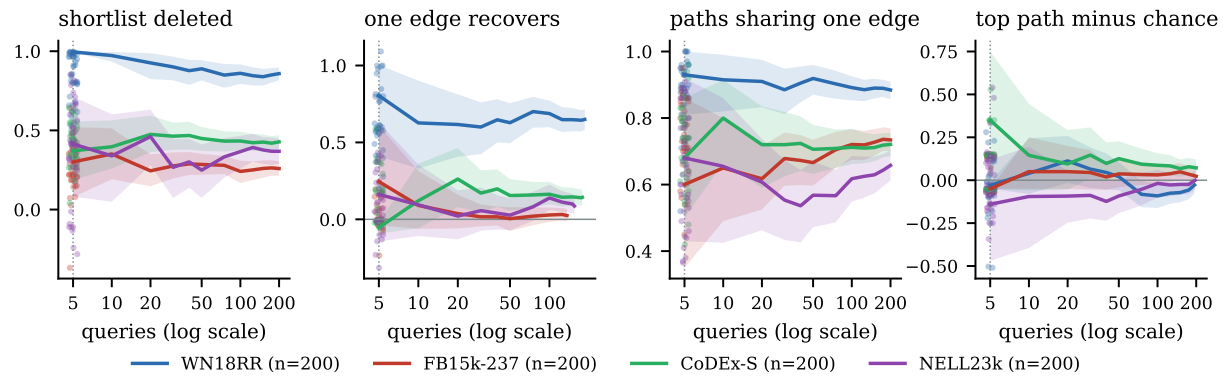


Figure 2. Each headline estimate against the number of queries, as one unbroken line with its 95% bootstrap band, from $n = 5$ to the full sample. The dots at $n = 5$ are every disjoint block of five queries, which is what a five-query benchmark could have reported.

Table 2. Main results. Shares are of each query's own starting probability. Every row uses all 200 queries per graph except the corrected row on FB15k-237 (100), integrated gradients (queries where τ is defined, 200 and 160) and the channel rows (100, 50, 200 and 200). Brackets are 95% bootstrap intervals over queries. "Chance" is the rate at which a random path from the pool would be the most damaging one, counting ties as hits.

	WN18RR	FB15k-237	CoDEx-S	NELL23k
set level: are the selected edges load-bearing?				
delete shortlist, share of p removed	0.858 [0.813, 0.898]	0.259 [0.212, 0.303]	0.427 [0.386, 0.470]	0.368 [0.290, 0.444]
delete as many random edges	0.000	0.000	0.000	0.007
ranking: does the order mean anything?				
Kendall τ, rank against effect	−0.009 [−0.054, 0.037]	0.048 [0.010, 0.085]	0.076 [0.043, 0.108]	0.041 [−0.023, 0.102]
corrected explanation	0.082 [0.036, 0.128]	0.153 [0.098, 0.208]	0.185 [0.146, 0.224]	0.103 [0.042, 0.160]
integrated gradients	n/a	n/a	0.489 [0.445, 0.531]	0.445 [0.372, 0.516]
top path is the most damaging	0.253	0.080	0.125	0.306
chance	0.280	0.057	0.053	0.307
relation channels, corrected gradient				
silence strongest explained channel	0.995	0.770	0.895	0.895
silence a random channel	0.020	0.000	0.002	0.101
share of effect present at half gain	0.29	0.23	0.19	0.03
gradient has the sign of silencing	59.2%	64.9%	60.8%	73.9%
strongest channel is the query relation	92.0%	85.7%	96.5%	71.5%
redundancy: many explanations or one?				
paths sharing the busiest edge	0.884	0.734	0.721	0.658
distinct edges / edges listed	0.248	0.484	0.512	0.295
that one edge recovers	0.650 [0.581, 0.716]	0.025 [−0.014, 0.066]	0.143 [0.093, 0.197]	0.086 [0.026, 0.147]
structure of the queries				
reverse fact already in graph	57.5%	0.0%	13.0%	4.5%
median tail degree	13	854	320	92

and from 0.030 to 0.111 on NELL23k, a gain of 0.081 [0.010, 0.151]; on the seen graphs the gain is 0.082 and 0.076. So the bug explains part of why the order was poor, but not most of it: the right gradient of the right model still ranks paths far from their measured effect. ULTRA's three-graph checkpoint, the model of the ULTRA paper, gives the same picture, with τ of 0.182 and 0.109, and it reproduces that paper's MRR, 0.472 against 0.472 on CoDEx-S and 0.241 against 0.239 on NELL23k (Table 1).

7.4. Integrated gradients rank better, but not well

The gradient is a slope at the point where every edge weight is one. Integrated gradients average that slope along the way from the empty graph to the full one [10]. Re-ranking the corrected explanation's own candidates by the integrated gradients summed over each path's edges raises τ from 0.185

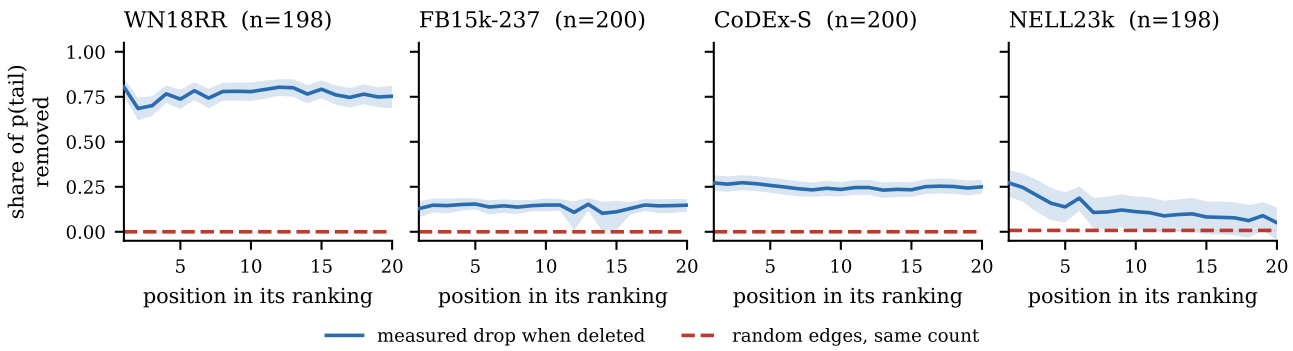


Figure 3. What the method ranks first is not what matters most. The horizontal axis walks down the method's own ordering of its 20 candidates; the vertical axis is the share of the gold tail's probability that deleting that path removes, with a 95% bootstrap band. A faithful ranking would fall from left to right. The dashed line deletes the same number of random edges.

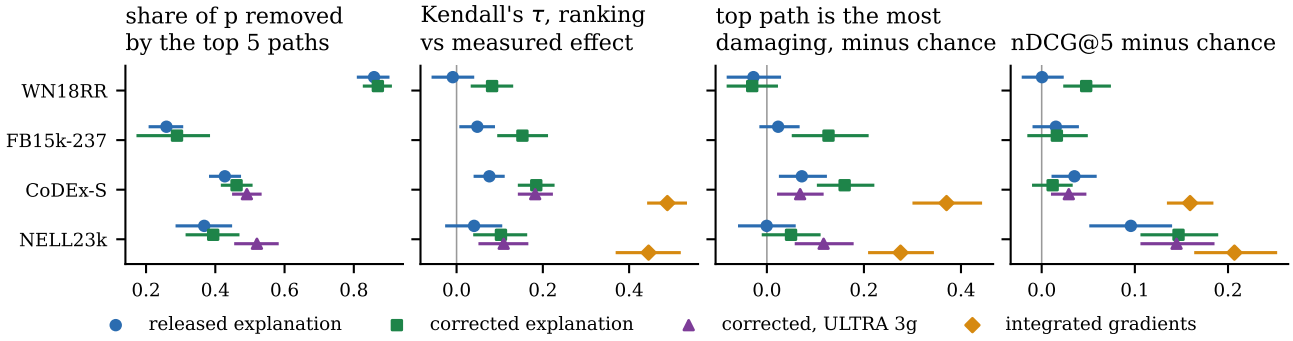


Figure 4. The headline numbers, one variable changed at a time: the released explanation, the direction-corrected one, the corrected one on ULTRA's three-graph checkpoint, and integrated gradients re-ranking the corrected explanation's own candidates. Points are means over queries, lines 95% bootstrap intervals. Integrated gradients was run on the held-out graphs only, and it re-ranks a fixed candidate set, so it has no joint-deletion value.

to 0.489 on CoDEx-S, a paired gain of 0.304 [0.257, 0.350], and from 0.140 to 0.445 on the 160 NELL23k queries where both are defined. How often the top path is the most damaging one, above chance, goes from 0.160 to 0.370 on CoDEx-S. Summed over a relation channel's edges, integrated gradients have the sign of that channel's silencing effect on 90.4% of the channels that matter, against 60.8% for the gradient, both averaged per query as in Table 2. It costs 16 backward passes instead of one, and with 16 steps the attributions add up to the total change only to within a median of 17% of it, so the integral is approximate. This is the largest improvement in the study, and it still leaves τ near 0.5 at best: the order of the paths is only loosely tied to what deleting them does.

7.5. Relation channels are switches, not dials

Turning a channel down shows why a slope misleads (Figure 5). Silencing the strongest channel the explanation uses removes 0.895 of the gold tail's probability on CoDEx-S and 0.895 on NELL23k, while a random channel removes 0.002 and 0.101 (Table 2). The response is not a dial. For channels whose silencing matters, only 0.19 of the full effect is present at half gain on CoDEx-S, where a straight line would give 0.5. 51.7% of them have less than a quarter of it, and 35.6% change direction somewhere along the dose, a non-monotone response like dopamine's inverted U [2]. Turning every explained channel down to half together removes nothing measurable on CoDEx-S (-0.027 [-0.071, 0.017]) and raises p on NELL23k (-0.181), against 0.909 at zero on CoDEx-S. WN18RR is the exception: there half gain on the explained channels already removes 0.489, about what a straight line gives. A slope taken at gain 1 cannot see a switch that sits near 0, and the corrected gradient has the sign of the silencing effect for only 60.8% of these channels on CoDEx-S and 73.9% on NELL23k, where chance is about half. The channel that matters most is usually the query's own relation, from 71.5% of queries on NELL23k to 96.5% on CoDEx-S. The paths do name that relation, as a hop; what they cannot show is that it acts as a switch. This is the Topic 8 result again, one level up: the gradient localises a channel, and localising it does not predict what editing it

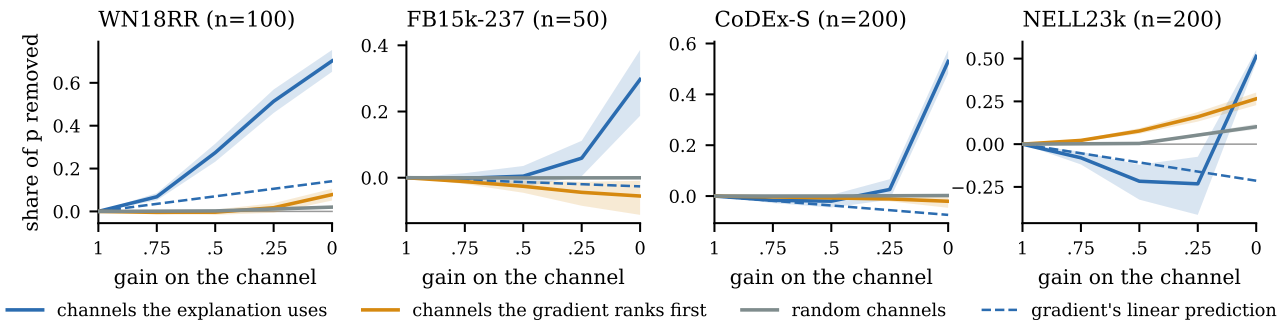


Figure 5. Turning one relation channel down from gain 1 to 0. Curves are the share of the gold tail's probability removed, averaged over queries, with 95% bootstrap bands, for the channels the explanation's paths use, the other channels the corrected gradient ranks highest, and random channels. The dashed line is the gradient's linear prediction for the explained channels.

Table 3. WN18RR split on whether the query's reverse fact is in the training graph. The forward edge is never present (0.0%).

	reverse present (115 queries)	reverse absent (85 queries)
paths sharing one edge	0.894	0.872
Kendall τ	−0.012	−0.003
top path is best	0.157	0.392
chance	0.217	0.372
one edge recovers	0.807	0.395
rewire, old loses	0.2808	0.0017

does.

7.6. One edge or many depends on the graph

The shortlist is presented as several independent paths. On WN18RR it is close to one. A single edge runs through 0.884 of the listed paths, only 0.248 of the listed edge slots are distinct, and deleting that one edge reproduces 0.650 of the effect of deleting the whole shortlist. The obvious objection is that these tails are poorly connected so that any deletion strands them; it fails, since the median tail degree is 13 and the result holds on tails of degree four or more (0.613).

The other graphs do not look like this. On FB15k-237 the busiest edge appears in 0.734 of paths and deleting it recovers 0.025 of the effect, an interval that includes zero; the paths there are genuinely different routes. The structural difference is in the graphs themselves (Tables 1 and 2): WN18RR is sparse, has 11 relation types and keeps symmetric relations, so 57.5% of its queries have their reverse fact in the graph, while FB15k-237 has 237 relation types, a median tail degree of 854, and 0.0% of queries with a reverse fact. The held-out graphs sit between the two and let the pattern be checked on graphs the model never saw.

7.7. The control I found by reading an example

While building a worked example on WN18RR, the top "path" came back as a single hop, and that hop was the stored reverse of a training fact. Table 4 shows the first such query in the benchmark's order. The model is asked what *sociology* is derivationally related to and answers *sociologist*, because the training graph already holds that fact the other way round. The explanation offers five paths. Two of them detour from sociology to *mores* and straight back before taking the same reverse hop, and deleting that one hop alone does everything that deleting all five paths does. This is not leakage, since the test edge itself is never in the graph, but it changes what there is to explain, so Table 3 splits WN18RR on it. Two findings survive the split: the paths funnel through one edge on both halves, and the ranking is no better than chance on both. What does not survive is how much that edge carries, 0.807 with the reverse fact present against 0.395 without, and almost all of the rewire effect. The first column is the sharpest thing in this paper. On those queries the model is doing a one-hop lookup, and the interface returns up to 20 paths of up to six hops. Nothing in that output says the answer came from one reverse lookup, which is the one thing a reader would most want to know.

7.8. Rerouting a path destroys rather than redirects

If the top path were the mechanism, pointing its last hop at another entity should carry the answer there. On WN18RR the old answer loses 0.1622 of probability while the new target gains 0.0282, and the effect comes almost entirely from the reverse-fact queries; elsewhere the rewire barely moves either (the old answer loses at most 0.0466 on NELL23k). Removing the hop takes away support without transferring it, which is MQuAKE's portability failure [13] in a setting where the edit is exact.

8. Discussion

The bug is the most concrete finding, and its lesson is general. A gradient explanation is a statement about whatever code was differentiated, and here that was not the code that makes the predictions. Nothing in the output looks wrong, and

Table 4. A worked example, the first query in the benchmark's fixed order that meets the rule in its heading, re-run under the benchmark protocol. Arrows show the direction of the stored fact, so a leftward arrow is a hop taken backwards along a reverse edge. Names are from the WN18RR text release.

WN18RR, first query with its reverse fact in the graph. Query: sociology − derivationally related form → ? answer **sociologist**, rank 1, $p = 0.196$	
path 1 (1 hop)	sociology ← derivationally related form − sociologist
path 2 (3 hops)	sociology ← synset domain topic of − mores
	mores − synset domain topic of → sociology
	sociology ← derivationally related form − sociologist
path 3 (5 hops)	sociology ← synset domain topic of − mores
	mores − synset domain topic of → sociology
	sociology ← synset domain topic of − mores
	mores − synset domain topic of → sociology
	sociology ← derivationally related form − sociologist
delete top 5 (6 edges)	p 0.196 → 0.000, rank 1 → 24859
delete busiest edge only	p 0.196 → 0.000, rank 1 → 24874: 100% of that effect
	that edge, in 4 of 5 paths: sociologist − derivationally related form → sociology

the deletion tests did not catch it. Checking the gradient against the model's own response did: the completeness of integrated gradients, then the scores and a finite difference.

The gradient finds a region of the graph the model depends on, and random deletion confirms it is not luck. As released it ranks little or nothing inside that region, and even fixed it ranks poorly; whether it decomposes the region into separate evidence depends on the graph. A heat map over edges and a ranked list of reasoning chains are different promises. NBFNet's gradient keeps the first and is presented as the second.

The redundancy result is the one I would hand to someone building on this. Ten paths that agree feel like ten witnesses. If nine are the same witness in different coats, a reader who spot-checks two and finds them sound has learned far less than they think, and the output looks the same whether that is happening or not. Five changes would help: run the explanation on the reversed graph; rank with integrated gradients, at 16 backward passes, if an order is going to be shown; report the busiest edge and confirm it with one deletion; print the distinct-edge count beside any list of paths; and flag queries whose reverse fact is already in the graph.

None of this says the model is broken. Its accuracy is in line with what the ULTRA paper reports, and the edges its gradient selects are load-bearing. What fails is the explanation layer: first a gradient taken through the wrong code path, and then, once that is fixed, a real but coarse signal over edges, served as a ranked list of discrete reasoning chains it does not support. The channel experiment suggests why a better gradient is not enough: much of the answer rides on relation channels that act as switches, which a slope cannot see.

9. Limitations

Overlapping paths. Deleting a path disturbs other paths that share its edges, so a single-path drop is not cleanly that path's. The redundancy rows of Table 2 measure this, and the set-level and single-edge numbers do not depend on it. **One model family**, four graphs: the claim is about this family's explanations, not path reasoning in general. **Correct predictions only**: what the explanation does on the model's mistakes is a separate question. **Smaller runs on the seen graphs**: the corrected benchmark has 100 queries on FB15k-237 and the channel experiment 100 on WN18RR and 50 on FB15k-237, against 200 elsewhere; integrated gradients and the second checkpoint were run on the held-out graphs only. **Integrated gradients with 16 steps**: completeness holds to a median of 17% of the total change on CoDEx-S; more steps would tighten it. **NBFNet-PyG**: its layer code has the same fallback, but I ran the checks on ULTRA only. **WN18RR node merges**: the release PyTorch Geometric loads merges 161 unrelated synsets (0.4%) into shared nodes; the checkpoint was trained on that graph, so I left it.

10. Conclusion

I tested whether the paths NBFNet reports are the paths it relies on, by editing a frozen checkpoint's graph rather than reading its gradients, on 800 queries over four graphs. The released explanation differentiates the model with its messages reversed; fixed, it matches the deployed model. Either way the selected edges are load-bearing, and the order they come in carries little: the fix and integrated gradients each improve it, and neither makes it faithful. Localising is not explaining, and only changing the graph tells the two apart.

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